

COMPUTER ARCHITECTURES

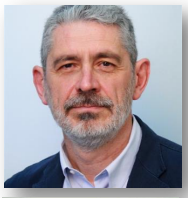
Turin Polytechnic University

Tashkent, Uzbekistan

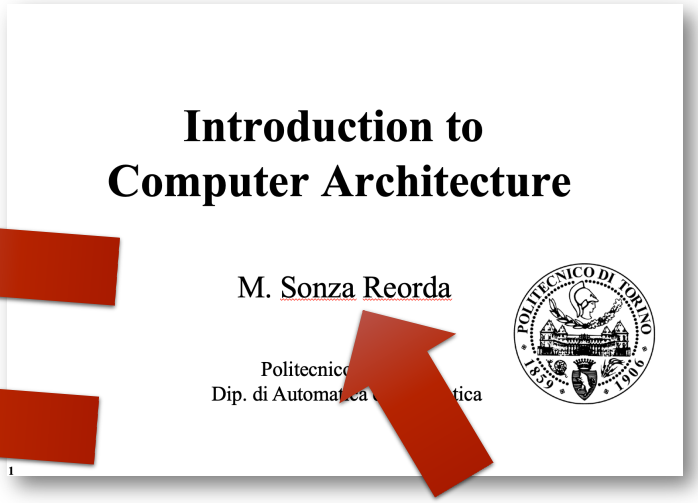


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Credits









2

Instructors

- **Giovanni Squillero**

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  `https://staff.polito.it/giovanni.squillero/at/`

- **Lobar Asretdinova**

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+ Telegram Chat

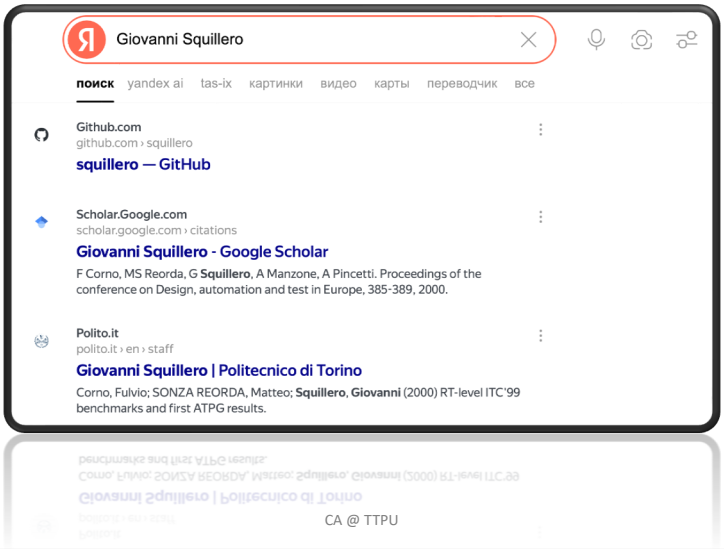
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Rule of engagement

- Conduct
- Attendance
- Exam

5

Who is this Squillero?



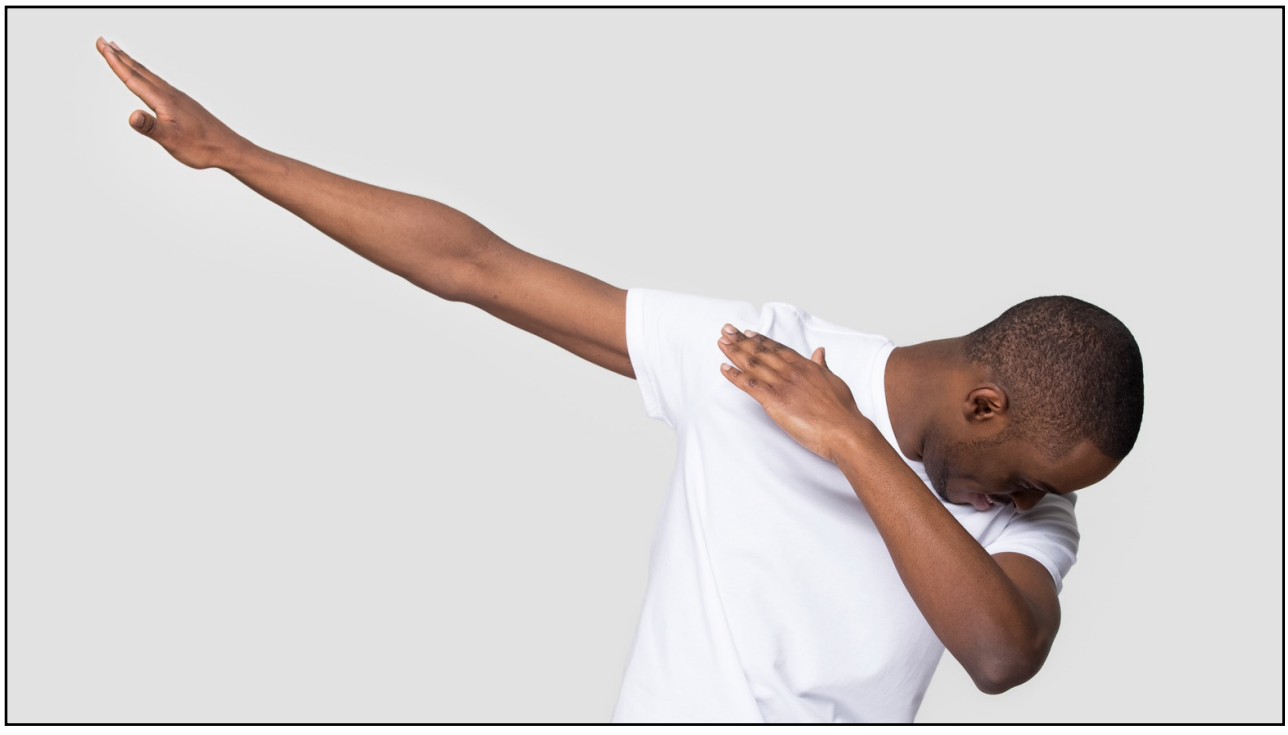
Why are we here?

- Software vs. Hardware
- Advanced programming
- Effective programming

Goals

- To introduce in an informal manner some basic concepts
- To describe the current situation
- To summarize how we reached the current state of the art in the area of computer systems

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Quick Comparison

Apple M5 Max

- Technology
  - 3nm Enhanced (TSMC N3P)
- Transistor Architecture
  - FinFET
- Clock speed
  - 4.4-4.6 GHz
- Core Configuration
  - 6 Super / 12 Efficiency
- Memory Bandwidth
  - Up to 614 GiB/s
- Max Power Draw
  - 80-100 W

Intel Core Ultra 9 290HX Plus

- Technology
  - 1.8nm Class (Intel 18A)
- Transistor Architecture
  - RibbonFET
- Clock speed
  - Up to 6.0 GHz
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About 15 atoms

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About 9 atoms

A silicon atom is about 0.2 nm wide

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Decade-long industry standard

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New Intel's GAA (Gate-All-Around)

Quick Comparison

1 step/clock tick?
Walk 100-150 times around the earth every seconds...

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Some 130,000 clock cycles in the flash of one single lightning bolt

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180,000 clock cycles in the flash of one single lightning bolt

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Typical workload?
Upgradeability?

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X Plus

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100-120 GiB/s

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Why programming in ASM!?



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Dario Amodei

CEO @ Anthropic

On 11 March 2025:

“In the next 3 to 6 months, AI is writing 90% of the code, and in 12 months, nearly all code may be generated by AI”



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Yann LeCun

Former Chief scientist @ META

Turing Award 2018

One “godfather of AI”

Contributions in NN

- CNN
- Autoencoders
- …

Creator of DjVu

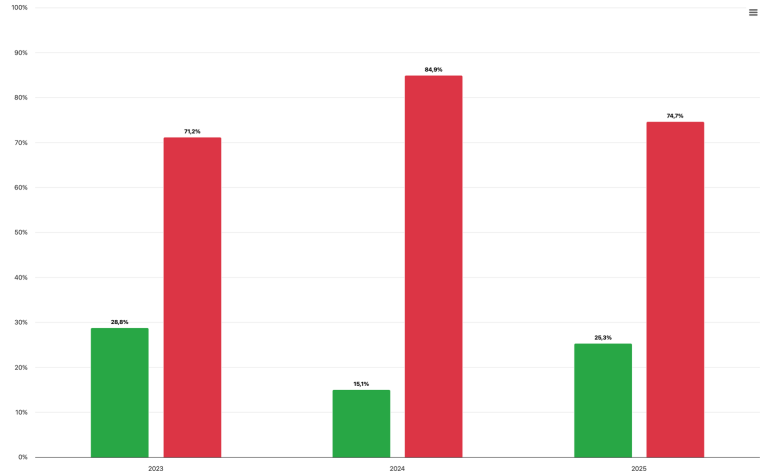
400k+ citations on Scholar

giovanni.squillero@polito.it



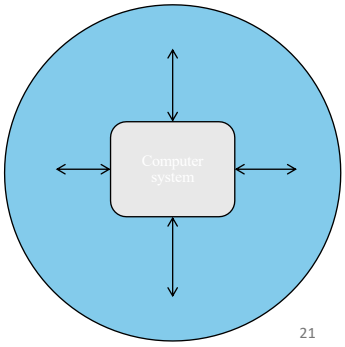
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Past years exams



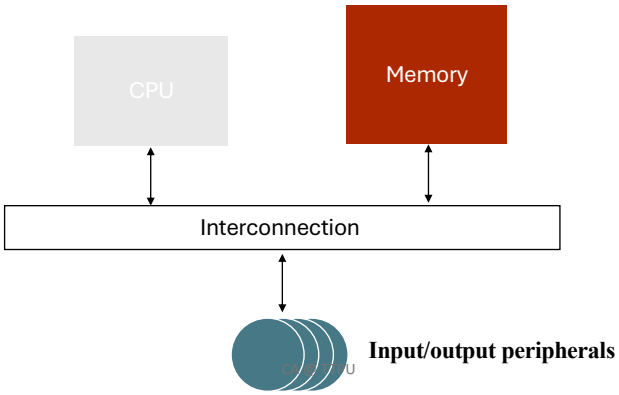
Computer systems

- A computer system is able to
- Receive inputs from the outside
- Perform some processing
- Send commands and data outside



Processor-based systems

- A computer system often corresponds to a processor-based system.



Advantages

- Computer systems have several advantages in terms of
  - Speed
  - Size
  - Dependability
  - Cost.

Key ideas

- Computer systems are based on
  - Techniques to represent information (numbers, text, multimedia)
  - Techniques to define the sequence of operations to be performed to solve a problem (i.e., algorithms)
  - Techniques to store data and automate operation execution.

Ages

- *Mechanic* age (until 1945)
- *Electronic* age (from 1945 to 1975) organized in 3 generations
- *VLSI* age (since 1975).

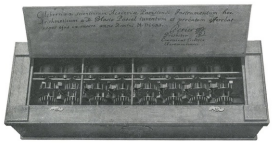
Mechanic age



- Abacus (2700-2300 BC)
- *B. Pascal* (1623-1662)
- *G. Leibniz* (1646-1716)
- *J. Jacquard* (1752-1834)
- *C. Babbage* (1792-1871)
- Z1 (1938), Z3 (1941) by *K. Zuse*
- Mark I (1944) and Mark II by *H. Aiken*

B. Pascal

- In 1642 he built a computing machine
- First working computer
- Able to perform additions and subtractions
- Included 2 sets of 6 toothed wheels each. Each wheel had 10 positions, corresponding to a decimal character
- Each set corresponded to a register, able to store a decimal value.



Leibniz

- Around 1671 he built a new, more powerful computing machine:
- Able to perform sum, subtraction, multiplication and division operations
- Composed of two Pascal machines (4 sets of wheels)
- Multiplication and division operations were performed by sequences of additions and subtractions.

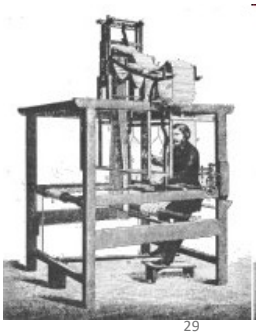
Jacquard loom

•It was built around 1801 by J. Jacquard.

•It is the first example of *programmable* machine, aimed at controlling the weaving process.

•The sequence of operations performed by the loom was decided by a set of pasteboard cards with punched holes

•Jacquard wrote a program composed of 24,000 cards to weave a tapestry with his image.



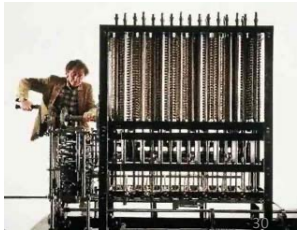
Differential machine

•It was designed by *C. Babbage* (the inventor of the tachymeter) around 1823.

•Its purpose was to compute tables of numbers to support sailors.

•It was able to execute a single algorithm (finite differences with polynomials).

•It included and output device based on engraving a copper plate with a steel tip.



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Analytical machine

- It is the evolution of the differential machine and was designed around 1834.
- It was composed of 4 parts:
  - The storage (i.e., the memory, composed of 1,000 words, each composed of 50 decimal characters)
  - The mill (i.e., the processing unit)
  - The input device (reading punched cards); cards defined the sequence of operations to be performed and the origin of the operands
  - The output device (copper plate or punched cards).

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Zuse

- Between 1939 and 1944 K. Zuse designed and manufactured in Germany 4 computer products (Z1, Z2, Z3, Z4).
- They are the first examples of programmable computers and are based on electromagnetic relays.
- For the first time a binary representation for numbers was used.
- All the Zuse prototypes were destroyed in Berlin during WWII.



Harvard Mark I e II

- They were manufactured (the former) and designed (the latter) by H. Aiken in Harvard, MA.
- The design was based on Babbage ideas, using relays. Storing was implemented through wheels.

Mark I

- Mark I was completed in 1944 in the IBM labs in Endicott.
- It owned 72 words of 23 decimal characters each, and a cycle time of 6 seconds.
- Input and output were based on tapes of punched paper.
- Problems:
  - Speed was limited by the movements of the composing parts
  - Information were transferred through levers and pulleys in a very unreliable manner.



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Turing

- When the first computing systems were being designed, in 1937 Alan Turing proposed a computer model able to execute a generic algorithm (Universal Machine).
- This model allows to
  - Distinguish what is computable and what is not
  - Identify the complexity of an algorithm.



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Colossus

- During WWII, Turing was part of the group of scientists who worked at Bletchley Park to develop techniques aimed at decoding the German messages encoded via the Enigma devices)
- In Bletchley Park the Colossus was built
- At the end of the war everything was destroyed and kept secret.



Electronic age

- Data flow correspond to movements of electrons
- Switching is fast thanks to vacuum tubes (also called *valves*).



ENIAC

- ENIAC (*Electronic Numerical Integrator and Calculator)* was built between 1943 and 1946 at the University of Pennsylvania by *J.W. Mauchly* and *J.P. Eckert*.
- It was composed of 18,000 valves and is weight was 30 tons.
- It was used to compute ballistic tables for the U.S. Army



Characteristics

- Decimal representation: each character was stored in a ring of 10 valves, where only one valve was on
- 20 accumulators, each storing 10 characters
- Special units for multiplication, division and square root
- Manual programming (via switches and cables)
- master programmer unit for multiple or iterative operations
- Special memories for constants (Function Tables)
- Data and code were separated
- Data input via card readers
- Data output via teletypewriters.

EDVAC

- EDVAC (*Electronic Discrete Variable Automatic Computer*) was designed in 1951 by the same team of ENIAC, including also *J. von Neumann* and *John Atanasof*.
- Characteristics:
  - Binary representation
  - Wider memory, divided in two parts:
    - Main memory: 1K word, lines of mercury technology
    - Secondary memory: 20Kword, coil technology
  - Code was also stored in memory
  - Data input/output via teletypewriters.

EDVAC

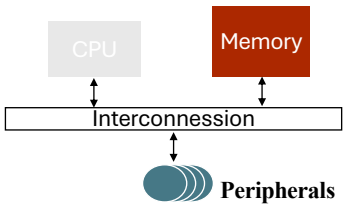
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EDVAC was never manufactured, because Eckert and Mauchley left the Università of Pennsylvania to fond *Unisys*.

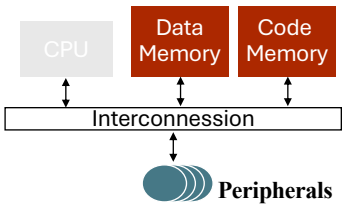
Von Neumann architecture

- We defined as von Neumann architecture the one based on a single memory storing both data and code, which is accessed by the CPU via a bus.
- The von Neumann architecture contrasts with the Harvard architecture, where the data memory is separated from the code memory.

Von Neumann architecture



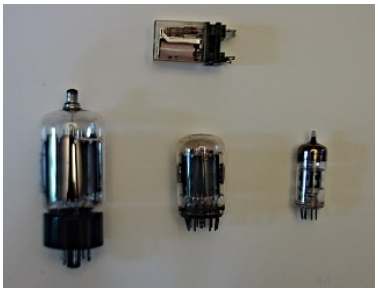
Von Neumann architecture



Harvard architecture

First generation

- It is composed of the computers built around 1950.
- Characteristics:
  - Technology based on valves
  - Centralized architecture: every operation is performed by the CPU
  - Machine language
  - No operating system
  - One user at a time can access the machine.



Second generation

- It is composed of the computers built around 1960.
- Characteristics:
  - transistors substitute valves
  - Cathodic ray and delay lines memory are substituted by those based on ferrite cores and magnetic drums
  - CPUs include index registers and hardware for floating-point operations
  - The first *hardware-independent* languages, such as ALGOL, COBOL and FORTRAN, are introduced
  - I/O processors free the CPU from the burden of controlling all the peripherals
  - First system software products (operating systems, compilers, libraries) are released

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n ferrite cores

ns

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eripherals

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The first transistor machine was the TX-0 (*Transistorized eXperimental computer 0*), built at MIT in the first 50s and then followed by TX-2.

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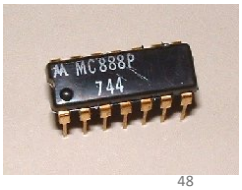
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Third generation

- Computers of the third generation appeared around 1965.
- Characteristics:
  - *Integrated Circuits* (ICs) substituted transistors
  - *Semiconductor memories* substitute the ferrite core ones
  - *Microprogramming* is introduced
  - *Pipelining* is introduced
  - *Operating systems* allow resource sharing in a system.

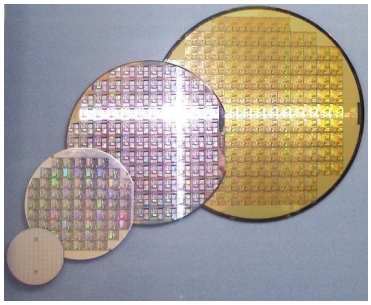


The VLSI age

- Integrating tens of thousands of transistors in a single circuit (*Very Large Scale of Integration*) makes it possible that entire systems (CPU, peripherals, memories) are manufactured in a single device.
- This provides significant advantages in terms of:
  - cost
  - size
  - speed.

Integrated circuits

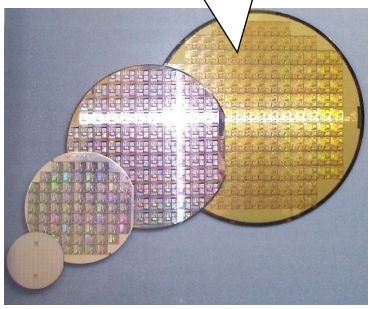
- An *Integrated Circuit* (or IC) is composed of:
  - silicon
  - package
  - pins
- Today, a single IC can include several millions (in some cases, billions) of transistors



Integrated circuits

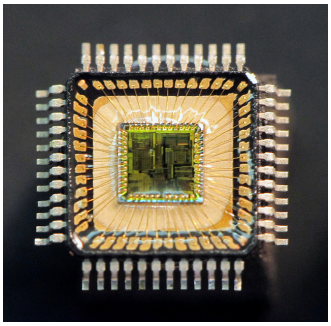
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Silicon *Wafer where* several *dies* are manufactured



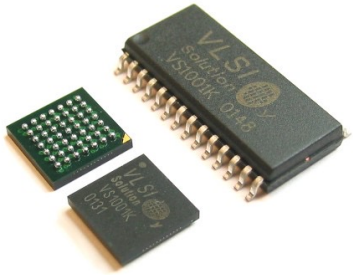
Bonding

- Once the different dies are cut from the wafer (slicing), each of them is inserted in its package and the input / output pads are connected to the pins (bonding).

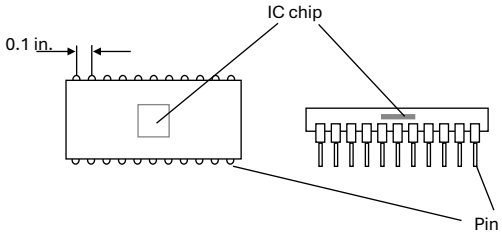


Package types

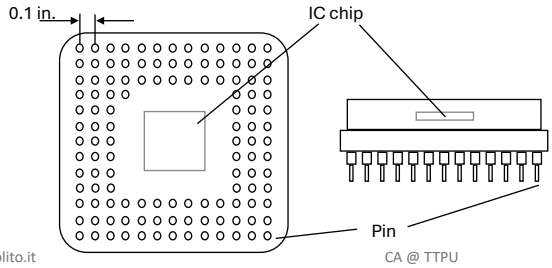
- The main ones include:
  - DIP (Dual In-Line Package): from 14 to 68 pins on 2 rows
  - PGA (Pin-Grid Array): 144 pins
  - BGA (Ball-Grid Array).



DIP and PGA



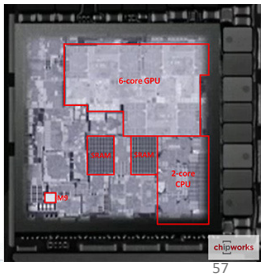
Dual In line Package
(DIP)



Pin Grid Array Package
(PGA)

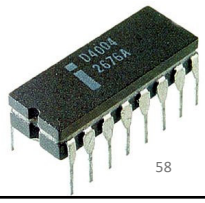
Integration density

- Thanks to the very high integration capabilities, today we can manufacture *System on a Chip* (SoC) devices, integrating in a single device one or more processors, memories, peripherals, special-purpose logic blocks.
- Example: the A9 chip by Apple



Microprocessors

- *Microprocessor* = processor manufactured on a single IC
- *Microcomputer* = computer based on a microprocessor
- The first commercial microprocessor was the Intel 4004 (1971).
- All processors until the late 80s belonged to the CISC (*Complex Instruction Set Computer*) category, and were characterized by a wide set of instructions (>100 instructions).
- CISC processors require a certain number of clock periods to execute each instruction.



RISC processors

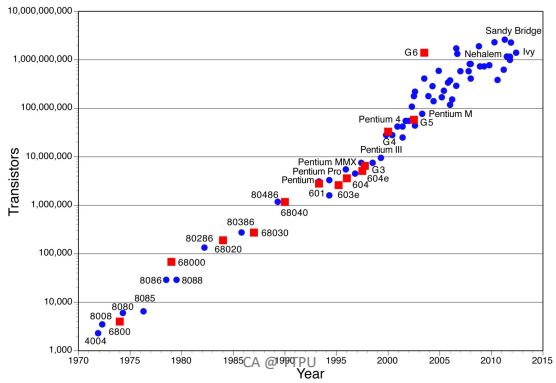
- RISC (Reduced Instruction Set Computer) processors are characterized by:
  - Reduced set of instructions (e.g., 32 instructions)
  - Pipelined architecture
  - High number of registers (e.g., 128).
- A RISC processor is typically able to complete the execution of an instruction at each clock period.
- The first RISC processors were released in the late 80s.

Superscalar processors

- Superscalar processors can complete the execution of more than one instruction per clock period
- They frequently use a set of RISC instructions
- Their architecture is based on a pipeline structure including multiple functional units.

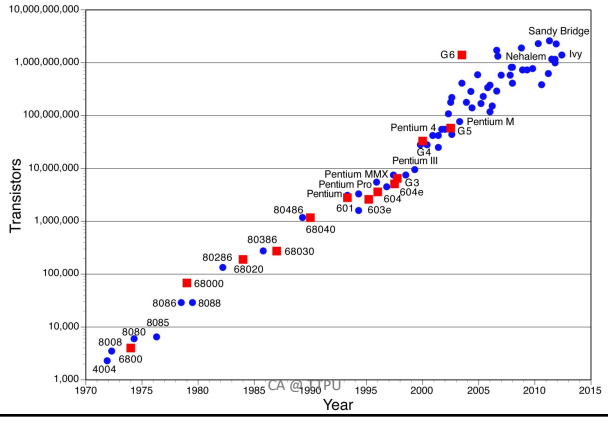
Moore law

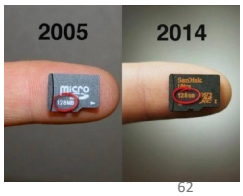
- It says that the number of transistors integrated on a single circuit doubles every 18/24 months.



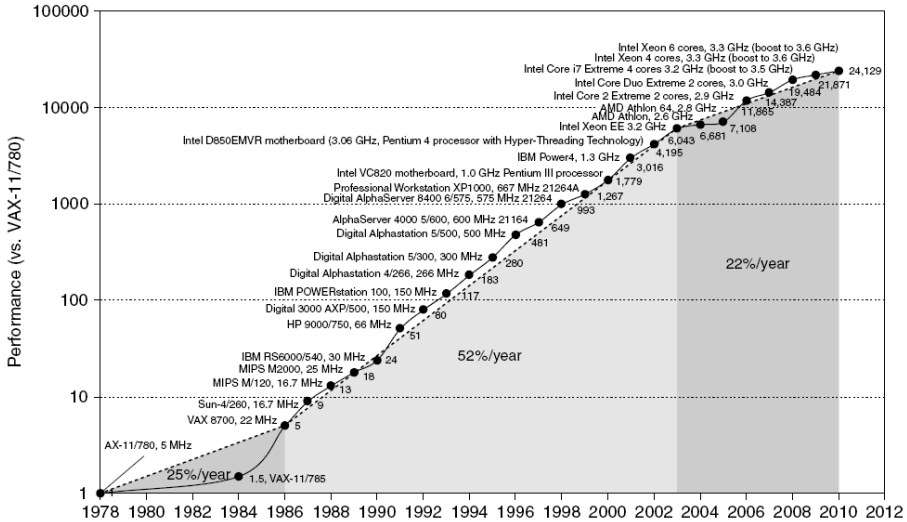
Moore law

- It says that the number of transistors integrated on a single circuit doubles every 18/24 months.

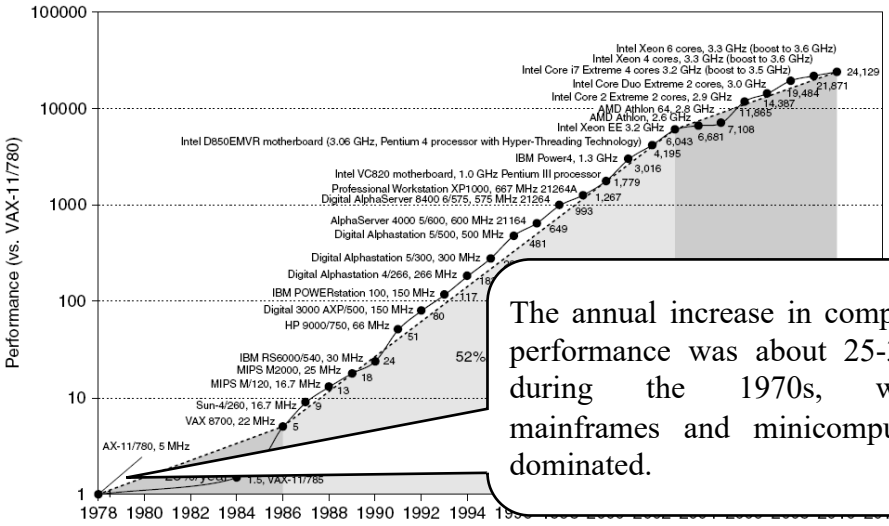




Microprocessor performance growth



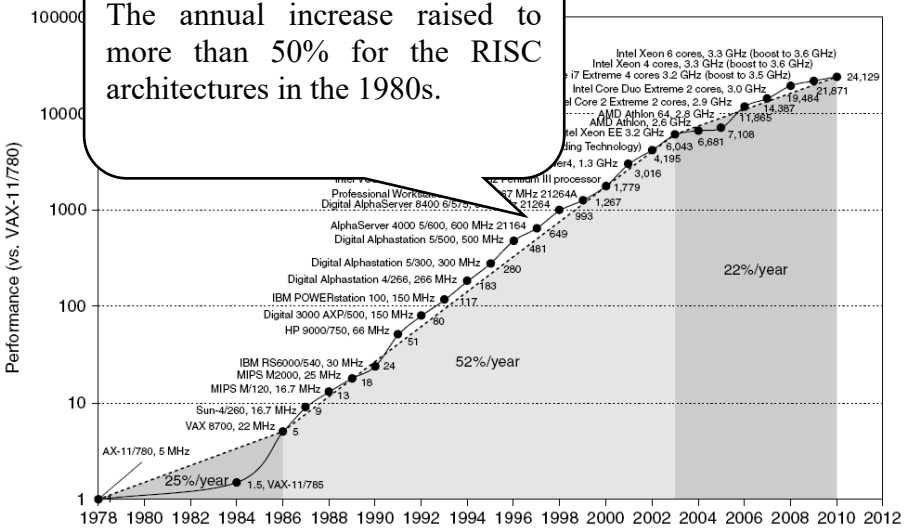
Microprocessor performance growth



The annual increase in computer performance was about 25-30% during the 1970s, when mainframes and minicomputers dominated.

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Microprocessor performance growth



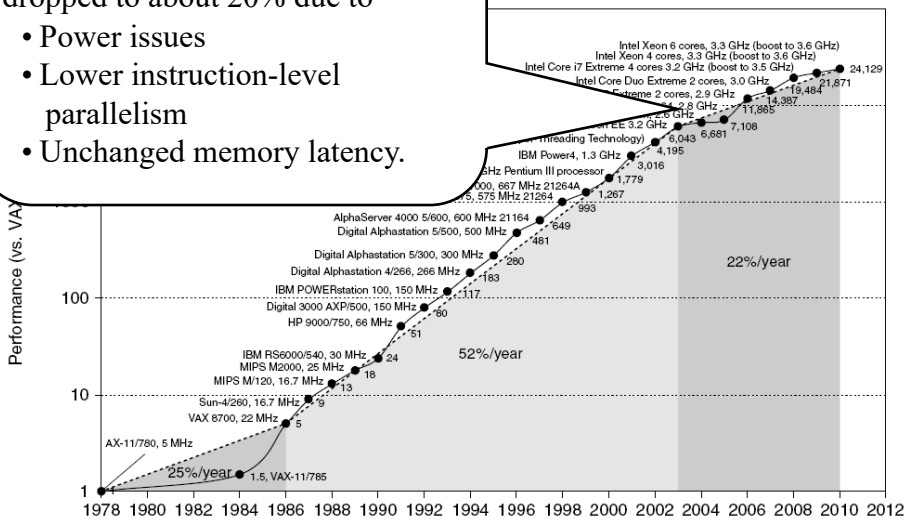
The annual increase raised to more than 50% for the RISC architectures in the 1980s.

65

Mi…ance growth

Starting from 2002, the yearly processor performance increase dropped to about 20% due to

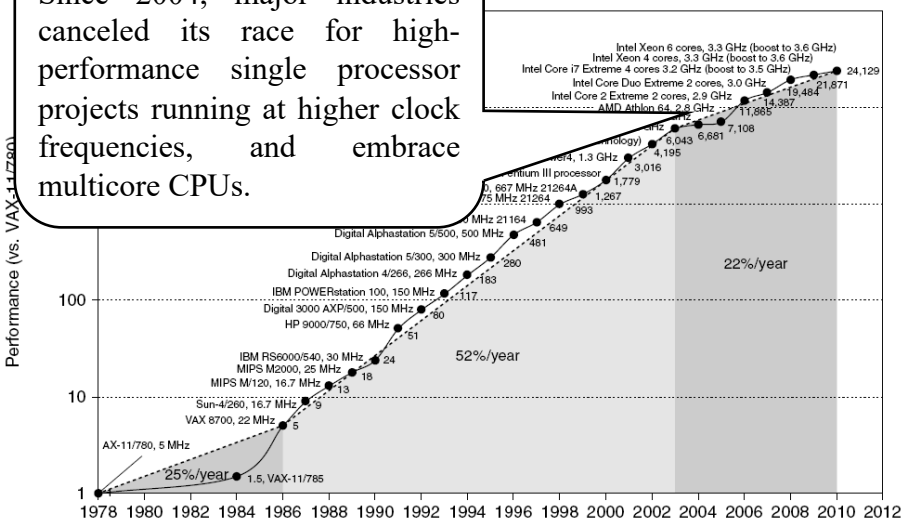
- Power issues
- Lower instruction-level parallelism
- Unchanged memory latency.



66

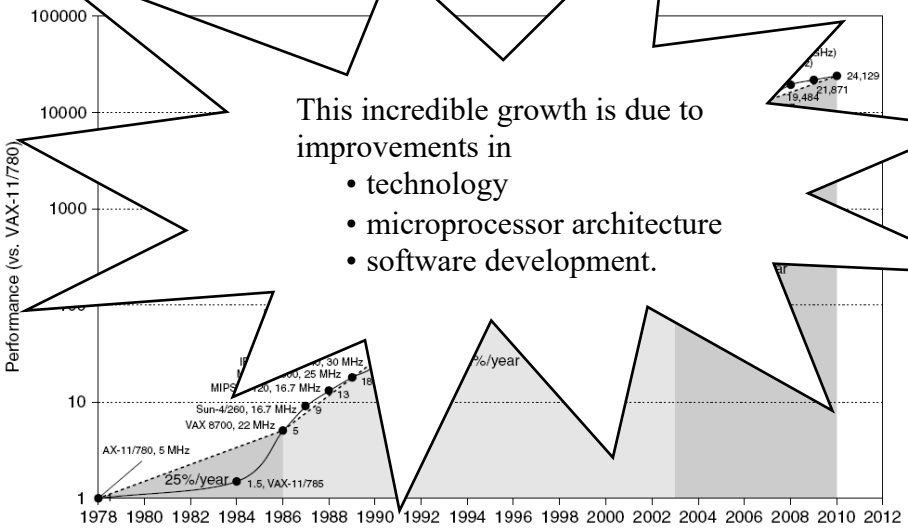
Micro…mance growth

Since 2004, major industries canceled its race for high-performance single processor projects running at higher clock frequencies, and embrace multicore CPUs.



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Microprocessor performance growth

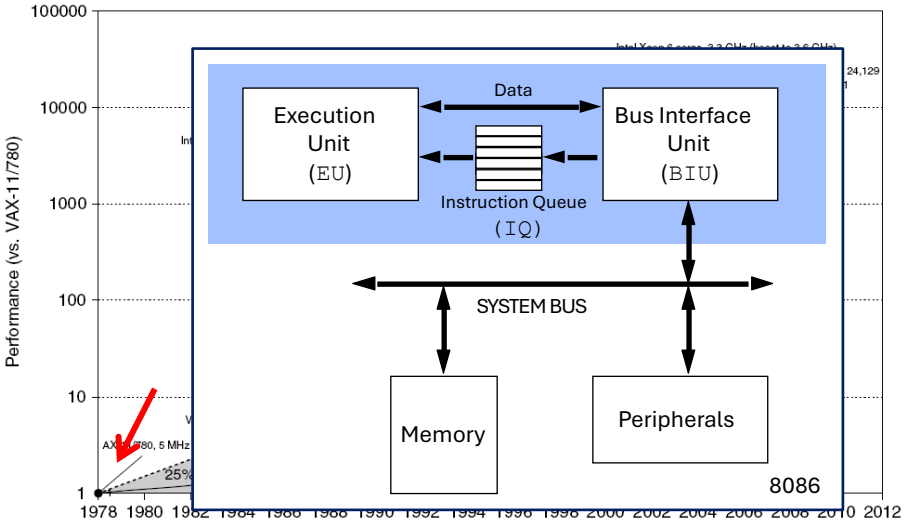


This incredible growth is due to improvements in

- technology
- microprocessor architecture
- software development.

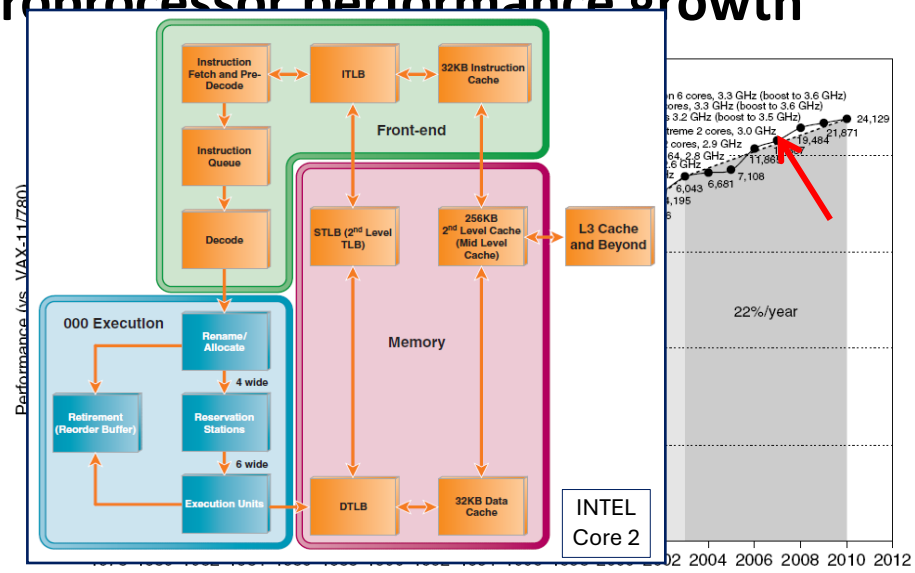
68

Microprocessor performance growth



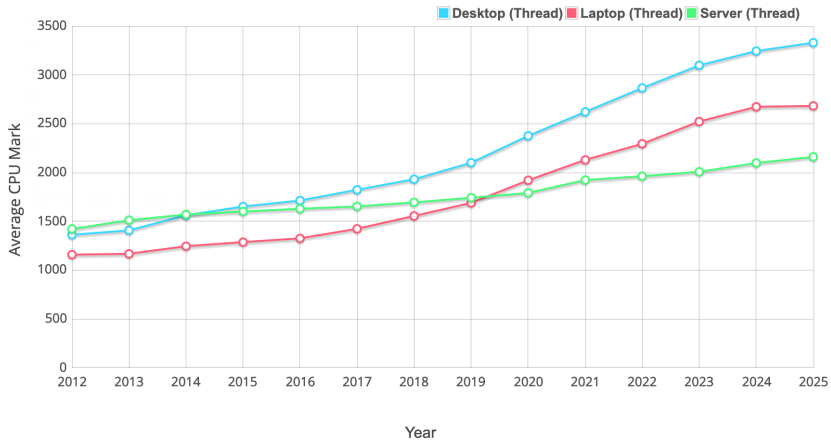
69

Microprocessor performance growth



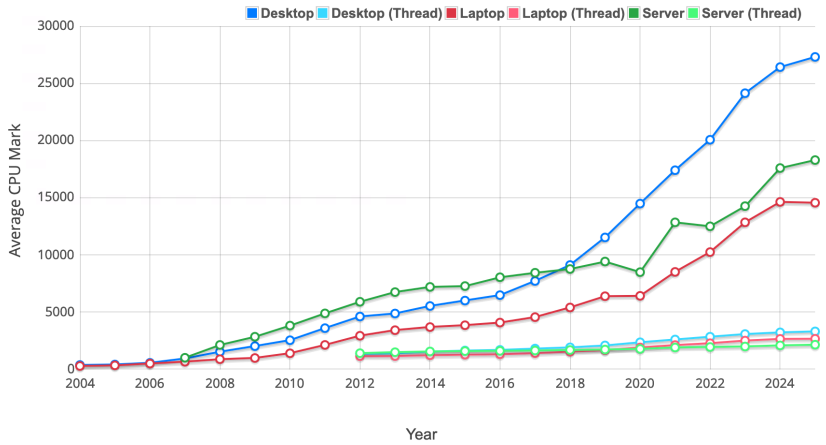
70

Moore Law (recent years)



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Moore Law (recent years)

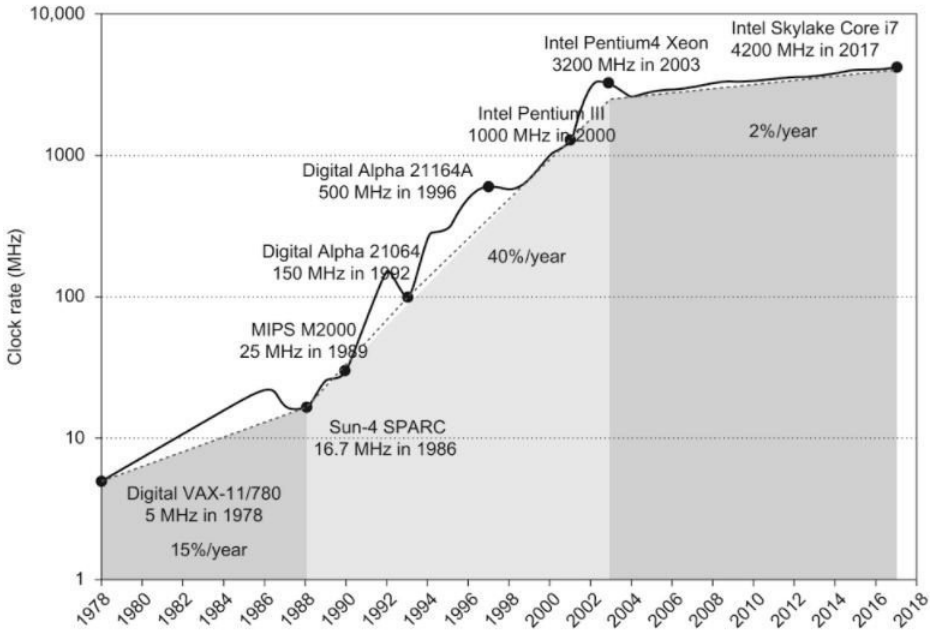


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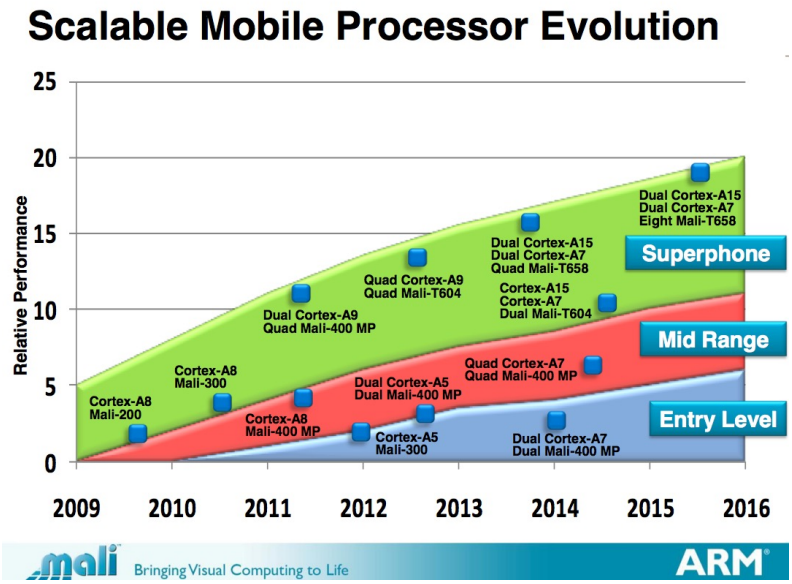
Trend

- The major players in the microprocessor market (Intel, AMD, IBM, ARM) are not investing anymore in the development of faster processors, but rather on multiprocessor single chip systems (i.e., multicore devices).

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10,000
1000
100
10
1
Clock rate (MHz)
1978
1980
1982
1984
1986
1988
1990
1992
1994
1996
1998
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
Digital VAX-11/780
5 MHz in 1978
15%/year
MIPS M2000
25 MHz in 1989
Sun-4 SPARC
16.7 MHz in 1986
Digital Alpha 21064
150 MHz in 1992
Digital Alpha 21164A
500 MHz in 1996
Intel Pentium III
1000 MHz in 2000
Intel Pentium4 Xeon
3200 MHz in 2003
Intel Skylake Core i7
4200 MHz in 2017
40%/year
2%/year

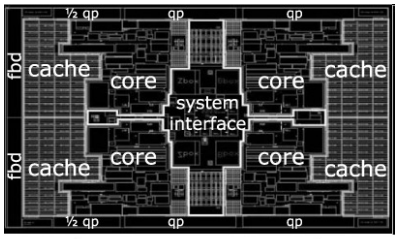
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Scalable Mobile Processor Evolution
Relative Performance
25
20
15
10
5
0
2009
2010
2011
2012
2013
2014
2015
2016
Cortex-A8
Mali-200
Cortex-A8
Mali-300
Cortex-A8
Mali-400 MP
Dual Cortex-A9
Quad Mali-400 MP
Cortex-A5
Mali-300
Dual Cortex-A5
Dual Mali-400 MP
Quad Cortex-A9
Quad Mali-T604
Dual Cortex-A7
Dual Mali-400 MP
Quad Cortex-A7
Quad Mali-400 MP
Cortex-A15
Cortex-A7
Dual Mali-T604
Dual Cortex-A15
Dual Cortex-A7
Quad Mali-T658
Dual Cortex-A15
Dual Cortex-A7
Eight Mali-T658
Superphone
Mid Range
Entry Level
mali
Bringing Visual Computing to Life
ARM

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Multicore processors

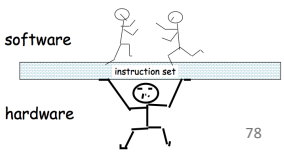
- Increase indefinitely the performance of a single processor is difficult
- Hence, over the last few years, multicore devices have been introduced, integrating multiple processors into a single device.
- Examples
- Cell (IBM, Sony, Toshiba)
- UltraSPARC T1 and T2 (Sun)
- Phenom (AMD)
- Core i7 (Intel)
- Tegra 3 (Nvidia).



	FET count	Voltage	Power
Core logic	430M	0.9-1.15V	100W
Sys Int	157M	0.9-1.15V	30W
L3 cache	1,420M	1.10V	20W
IO logic	39M	1.10V	20W
Chip Total	**2.046B**		**170W**

Instruction Set Architecture

- Processors are organized in families
- Processors of the same family are compatible at the software level, that is
- The code of any application developed for a processor can be run on all subsequent processors in the same family
- Compatibility is obtained by adopting the same Instruction Set Architecture (ISA), corresponding to the processor information set used by the programmer.



Microcontrollers (or MCUs)

- These devices are built on a single IC and each targets some specific special-purpose application.
- They are composed of:
- processor (with very variable features depending on the application)
- memory ROM, RAM, Flash modules
- ports and interfaces for I / O
- counters.
- In the field of microcontrollers performance is as important as other parameters, such as cost, consumption, size, reliability, and ease of use.

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MCU market

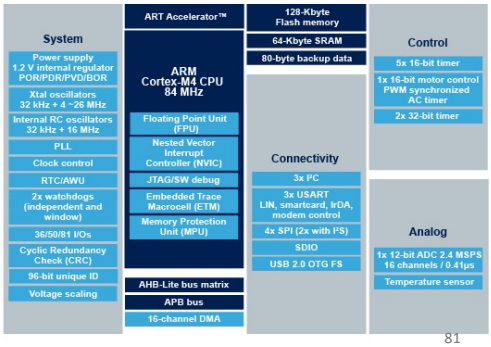
	2012	2013	2014	2015	2016	2017	CAGR
Total Semiconductor	325,367	339,666	361,612	385,052	395,974	413,602	4.9%
Microcontroller (MCU)	16,008	16,202	17,211	18,799	19,307	20,480	5.1%
4 bit MCU	154	159	161	157	145	133	-2.8%
8 bit MCU	6,057	6,565	6,936	7,532	7,768	8,259	6.4%
16 bit MCU	4,021	3,611	3,765	4,060	4,053	4,019	0.0%
32 bit MCU	5,776	5,868	6,349	7,050	7,341	8,069	6.9%

All figures are in millions of USD
CAGR = Compound Annual Growth Rate

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STM32

- It is a family of MCUs with 32-bit CPUs designed and manufactured by STMicroelectronics
- The first family products were introduced in 2006
- Family devices are used in many applications, such as
- Automotive
- Industrial
- Medical
- Railway ...
- More than one billion devices of this family have been sold already.



Special-Purpose systems

- These systems are designed and built to run a single application
- They are often inserted into more complex systems, which may include mechanical parts (mechatronic systems).

Special-Purpose system design

•There are several alternatives when designing a special-purpose system:

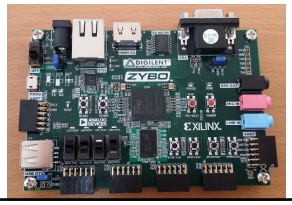
- SW Solution:
  - Use an existing board (equipped with a microprocessor or microcontroller that runs a program, usually contained in a ROM or Flash)
- HW Solution:
  - Based on an *ASIC* (*Application Specific IC*) designed and implemented specifically for that application.
  - If an ASIC contains one or more processors, it is called *SoC*.

Software solution

•It is so defined because it often uses commercial hardware, based on a microprocessor (or microcontroller) board

•In this case, the implementation work is focused on writing the software executed by the processor

•It is typically the least efficient solution in terms of execution speed, but also the least
expensive and more flexible.

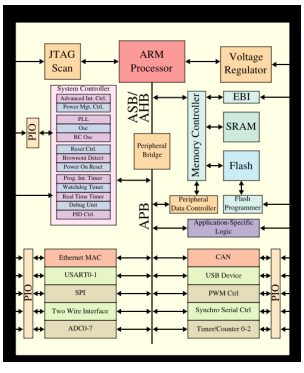


Hardware solution

- It consists in realizing an ASIC that implements the desired functionality
- The cost is much higher in terms of design and production, but the unit cost of the finished product is much lower
- It makes sense when the volumes are high (> 1 million)
- Moreover (in general)
- the performance is higher
- the footprint is smaller
- consumption is minimized
- flexibility is much smaller: once designed and built, the ASIC is no longer modifiable.

System on Chip

- Many applications (such as mobile phones) are based on complex ICs (called Systems on Chip or SoCs)
- They integrate
- one or more processors
- various memory modules
- the interface modules
- any specific modules
- Each SoC is aimed at a specific application and integrates the necessary IP modules.



Infineon Airbag SoC

